

외계행성 대기

Exoplanetary Atmospheres

Lecture 5: Radiative Transfer II — Solutions and the Two-Stream Approximation

2026 October 5 (Monday)

Based on: Chapter 6 of Seager, *Exoplanet Atmospheres* (2010), and Chapter 3 of Heng, *Exoplanetary Atmospheres: Theoretical Concepts and Foundations* (2017)

선광일 (Kwangil Seon)
KASI / UST

Outline – Lecture 5

The Emergent Spectrum: A Conceptual Picture

An Introduction to Line Formation

Solutions I: Transmission and Thermal Emission

Solutions II: Scattering Atmospheres

The Two-Stream Approximation

The Diffusion Approximation

Clouds and Monte Carlo Radiative Transfer

Summary

Lecture 4 built the tools: optical depth, opacity, LTE, the source function, and the transfer equation with its formal solution. Today we *use* them. Two questions organize the lecture:

1. **What does a spectrum tell us?** Why lines appear in absorption or in emission, at which depth they form, and what they reveal about the vertical temperature structure (Seager Ch. 6).
2. **How do we actually solve the equation** when scattering is present and the angular dependence cannot be carried along in full? The answer used in nearly every exoplanet climate and retrieval code is the **two-stream approximation** (Heng Ch. 3).

Working assumptions throughout: a one-dimensional, time-independent, plane-parallel atmosphere made of homogeneous layers, in hydrostatic equilibrium, with a *known* temperature–pressure profile. How that profile is itself determined is the subject of the next lecture (Heng Ch. 4, Seager Ch. 9).

The equation we solve, and its sign conventions

Along a ray, the intensity loses κI and gains the emission coefficient ε . With z the height and $\mu = \cos \theta$ measured from the upward normal ($dz/ds = \mu$), Seager's Eq. (6.1) reads

$$\mu \frac{dI(z, \mu, \nu)}{dz} = -\kappa(z, \nu) I(z, \mu, \nu) + \varepsilon(z, \mu, \nu).$$

Measuring the optical depth *downward* from the top, $d\tau_\nu = -\kappa dz$, and writing $S = \varepsilon/\kappa$ turns this into the form used by both books:

$$\mu \frac{dI}{d\tau_\nu} = I - S$$

- ▶ $\mu > 0$: outgoing (upward) rays; $\mu < 0$: incoming (downward) rays. $\tau_\nu = 0$ at the top of the atmosphere.
- ▶ All quantities are monochromatic; the subscript ν is dropped when no confusion arises.
- ▶ Everything in this lecture is about choosing S and solving this one line.

A planetary spectrum has two peaks

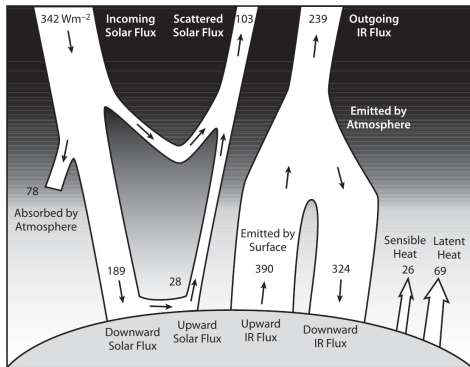


Figure 1: Energy balance of Earth's atmosphere (global means, in W m^{-2}). Seager (2010), Fig. 6.1.

- ▶ A **visible peak** of scattered starlight and an **infrared peak** of thermal emission.
- ▶ Thermal emission has two sources: absorbed stellar energy that is reradiated, and *interior* energy (residual heat of formation, radioactive decay). For hot Jupiters the first overwhelms the second.
- ▶ Reading the figure: of 342 W m^{-2} incident, 103 are scattered back ($A_B = 0.30$) and 239 leave as infrared, i.e. $\sigma T^4 = 239$ gives $T = 255 \text{ K} = T_{\text{eq}}$.
- ▶ The surface emits 390 W m^{-2} ($= \sigma 288^4$) because the atmosphere returns 324 W m^{-2} downward: the greenhouse effect, written in fluxes.

Continuum, absorption lines, emission lines

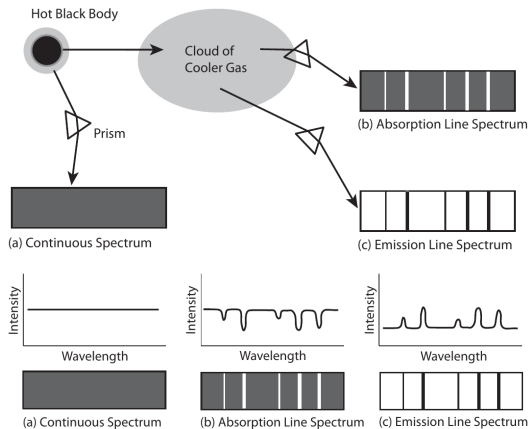


Figure 2: Origin of the three kinds of spectra (top) and what they look like over a narrow wavelength range (bottom). Seager (2010), Figs. 6.2 and 6.3.

- **Continuous spectrum:** a black body; the spectrum depends only on its temperature.
- **Absorption lines:** a cooler gas in front of the hot black body takes “bites” out of the continuum at the wavelengths its atoms and molecules absorb. The absorbed energy is re-emitted isotropically, so little of it returns to the line of sight.
- **Emission lines:** the same gas seen *without* the hot background. Only the re-emitted photons reach us, and they appear at exactly those wavelengths.
- What matters is not the gas alone but the gas *compared with what lies behind it*.

A layered atmosphere: the temperature gradient decides

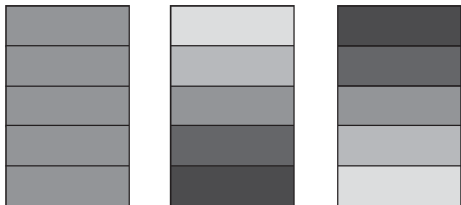


Figure 3: A five-layer atmosphere; darker means hotter. Left: isothermal. Middle: hot at depth, cooler above. Right: a temperature inversion. Seager (2010), Fig. 6.4.

Treat each layer as a homogeneous slab with its own temperature and the deepest layer as a black body.

- ▶ **Temperature decreasing outward:** every layer is a cool cloud in front of a hotter one $\Rightarrow$ an **absorption spectrum**. Each layer adds its own bites.
- ▶ **Temperature inversion** (hotter upward) $\Rightarrow$ an **emission spectrum**.
- ▶ **Isothermal:** no cooler gas in front, no hotter gas behind $\Rightarrow$ a black-body continuum with *no* features at all, however large the opacity.

The last point is counter-intuitive and worth proving (next section): spectral features measure temperature *contrasts*, not the mere presence of absorbers.

Transmission and scattered-light spectra

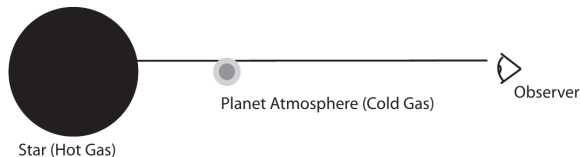


Figure 4: Origin of a transit transmission spectrum: hot stellar light through the cool planetary atmosphere. Seager (2010), Fig. 6.5.

- ▶ **Transmission:** the star is the hot black body, the planet's limb is the cool cloud $\Rightarrow$ always an absorption spectrum. It is isolated by dividing the in-transit spectrum by the out-of-transit stellar spectrum.
- ▶ **Scattered light** (visible, near-IR): Seager prefers “scattered” to “reflected”, because the light is refracted, diffracted and Rayleigh-scattered rather than bounced off a surface. To first order Earth's visible spectrum is sunlight that crossed the atmosphere *twice* (down to the ground or a cloud deck and back up), a weighted mix of clear and cloudy paths.

The isothermal atmosphere, layer by layer

Take an n -layer LTE atmosphere whose deepest layer n is optically thick, so $I_n = B_n$. With Kirchhoff's law $\epsilon_n = \kappa_n B_n$,

$$\mu \frac{dI_n}{dz} = \epsilon_n - \kappa_n I_n = \kappa_n (B_n - I_n) = 0.$$

The intensity entering layer $n - 1$ is therefore B_n , and in that layer

$$\mu \frac{dI_{n-1}}{dz} = \kappa_{n-1} (B_{n-1} - B_n) = 0 \quad \text{because } T_{n-1} = T_n.$$

Repeating up to the top, $I_0 = \dots = I_{n-1} = I_n = B$.

- ▶ Absorption and re-emission still happen in every layer, but they occur at the *same temperature* and hence at the same wavelengths: every photon removed is replaced by one of the same frequency.
- ▶ Note what the proof used: only the equality of temperatures, not the size of κ_ν . A strongly absorbing and a weakly absorbing wavelength give the same $I = B$.

The isothermal atmosphere, from the formal solution

For an emitting atmosphere without scattering and with no radiation entering from above, the emergent intensity is (Lecture 4)

$$I(0, \mu, \nu) = \int_0^\infty B(T(\tau_\nu), \nu) e^{-\tau_\nu/\mu} \frac{d\tau_\nu}{\mu}.$$

If T is constant, B leaves the integral, and $\int_0^\infty e^{-\tau/\mu} d\tau/\mu = 1$, so

$$I(0, \mu, \nu) = B(T, \nu), \quad F(0, \nu) = \pi B(T, \nu).$$

- ▶ The kernel $e^{-\tau/\mu}/\mu$ is a *normalized weighting function*: the emergent intensity is a weighted average of B over depth, with weights concentrated near $\tau \sim \mu$.
- ▶ A weighted average of a constant is that constant. Only when B varies with depth can different wavelengths, weighted over different depths, come out different.

A temperature gradient creates lines

Compare a wavelength in the line core, ν_l , with one in the nearby continuum, ν_c . By definition $\kappa(\nu_l) > \kappa(\nu_c)$.

- ▶ From $d\tau_\nu = \kappa_\nu dz$ (depth measured downward), the same value $\tau_\nu \approx 1$ is reached *higher up* in the line core than in the continuum: $z_l > z_c$.
- ▶ The emergent intensity is essentially B at $\tau_\nu \approx \mu$ (Eddington–Barbier). So the line core samples higher layers, the continuum deeper ones.
- ▶ **Temperature decreasing outward:** $T(z_l) < T(z_c)$, so $I(\nu_l) < I(\nu_c)$ — an **absorption line**.
- ▶ **Temperature increasing outward (inversion):** $T(z_l) > T(z_c)$, so $I(\nu_l) > I(\nu_c)$ — an **emission line**.

A spectrum across a line is therefore a *scan in altitude*: the wings, with lower opacity, look deeper, and the core looks highest.¹

¹Seager's text at this point reads " $I(\tau_\nu, \mu, \nu_l) < I(\tau_\nu, \mu, \nu_l)$ "; the right-hand side should be the continuum, ν_c .

Line formation in a toy atmosphere

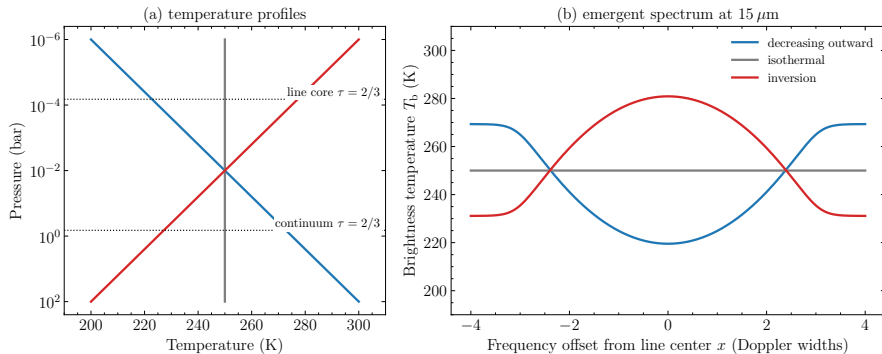


Figure 5: A line with a Gaussian opacity profile (peak 10^4 times the continuum) in three LTE atmospheres. Left: the temperature profiles and the pressures where $\tau = 2/3$ in the continuum and at line center. Right: the emergent flux as a brightness temperature. Computed with $F = 2\pi \int B E_2(\tau_\nu) d\tau_\nu$; script `scripts/make_line_formation.py`.

The continuum reports T near 0.7 bar, the line core T near 10^{-4} bar; the isothermal case is featureless despite the huge opacity contrast.

The emergent flux as a depth-weighted Planck function

Integrating the emergent intensity over the outgoing hemisphere,

$$F(0, \nu) = 2\pi \int_0^1 \int_0^\infty B(\tau_\nu, \nu) e^{-\tau_\nu/\mu} d\tau_\nu d\mu = 2\pi \int_0^\infty B(\tau_\nu, \nu) E_2(\tau_\nu) d\tau_\nu,$$

using $\int_0^1 e^{-\tau/\mu} d\mu = E_2(\tau)$, the exponential integral $E_n(x) = \int_1^\infty t^{-n} e^{-xt} dt$ (Seager Eq. 6.26).

- ▶ $2E_2(\tau)$ is the **flux weighting function**: it is normalized ($\int_0^\infty 2E_2 d\tau = 1$), equals 2 at the top and falls roughly as $e^{-\tau}/\tau$ at depth.
- ▶ This is the working formula for thermal emission from an exoplanet without scattering: a black body in each layer plus the (much harder) opacities. Most retrieval codes evaluate exactly this sum.
- ▶ Written per unit $\log p$, the integrand $B E_2 d\tau/d \log p$ is the **contribution function**: it shows which pressures a given wavelength actually probes.

Why a band has a floor: the minimum flux

Earth's mid-infrared spectrum (Lecture 1, Seager Fig. 3.4) shows the $15\text{ }\mu\text{m}$ CO_2 band: its bottom follows a 215 K black body, not zero.

- ▶ At a **low-opacity** wavelength the photons escape from deep, warm layers: this is the **maximum** flux, set by the surface or deep temperature.
- ▶ At a **high-opacity** wavelength the radiation is absorbed and re-emitted layer after layer; what finally escapes was emitted near the top: this is the **minimum** flux, set by the coldest layer the band can reach.
- ▶ The band depth therefore measures the *temperature range* of the atmosphere between the continuum and the band-center photospheres, not the amount of CO_2 as such.

In an LTE atmosphere the flux at any wavelength is never zero; it is the Planck function of whatever layer that wavelength “sees”. Hence brightness temperatures are a direct, if coarse, thermometer at different heights.

Reading Earth's temperature profile from its spectrum

- ▶ The deep O_3 ($9.6\ \mu\text{m}$) and CO_2 ($15\ \mu\text{m}$) absorption bands tell us that temperature decreases with height where they form: the **troposphere**, which holds most of the mass.
- ▶ At the very center of both bands there is a **narrow emission spike**. The core forms highest, so somewhere above the troposphere the temperature must rise again: the **stratospheric inversion**, heated by ozone absorbing solar UV.
- ▶ Why is the spike narrow? The opacity of a band scales with the number density of absorbers. Only at the band center is the opacity large enough for $\tau \approx 1$ to be reached in the low-density stratosphere; a little off-center, $\tau \approx 1$ drops back into the troposphere.

The width of an emission core thus encodes how high, and in how thin a layer, the inversion sits.

Thermal inversions in hot Jupiters?

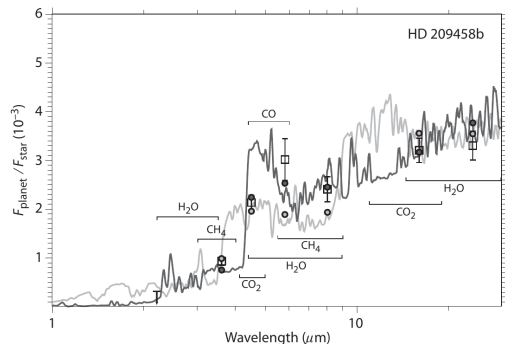


Figure 6: HD 209458b, Spitzer secondary-eclipse photometry (squares) with a model with absorption features only (gray) and one with a deep inversion (black). Seager (2010), Fig. 6.6.

- ▶ Tidally locked hot Jupiters show a permanent dayside, observed at secondary eclipse.
- ▶ Some appeared to show H_2O in *emission* with no absorption bands at all. By the argument used for Earth, that requires an inversion extending *down* to the layers where the bands form, i.e. a deep dayside inversion.
- ▶ Later data changed the case of HD 209458b: HST/WFC3 spectroscopy showed water in *absorption* and no inversion (Line et al. 2016). Inversions do exist in the hottest planets, e.g. WASP-121b (Evans et al. 2017), where TiO/VO or atomic metals can absorb starlight high up.
- ▶ The physics of the diagnosis is unchanged; only the data improved.

Pure transmission: Beer's law

In transmission there is neither emission nor scattering *into* the beam: $\varepsilon = 0$, hence $S = 0$, and

$$\mu \frac{dI}{dz} = -\kappa I \quad \implies \quad I(z_f) = I(z_i) \exp \left[-\frac{1}{\mu} \int_{z_i}^{z_f} \kappa dz' \right], \quad I(\tau_v) = I(0) e^{-\tau_v/\mu}.$$

In a transit the relevant quantity is the ratio of transmitted to incident stellar light along one ray, the stellar intensity cancelling:

$$\frac{F_{S,p,\text{trans}}(\nu)}{F_{S,*,\text{trans}}(\nu)} = e^{-\tau_v},$$

where τ_v is now the optical depth *along the chord* through the limb (Seager Eq. 6.21).

- This is one ray. The observed transit depth sums rays at all impact parameters through the annulus, each with its own chord optical depth (Lecture 4: $\tau_{\text{chord}} \approx \sqrt{2\pi R_p/H}$ times the vertical one).
- Pressure broadening matters at depth, which is why the sum over altitudes cannot be skipped in a real calculation.

A transmission spectrum: HD 189733b

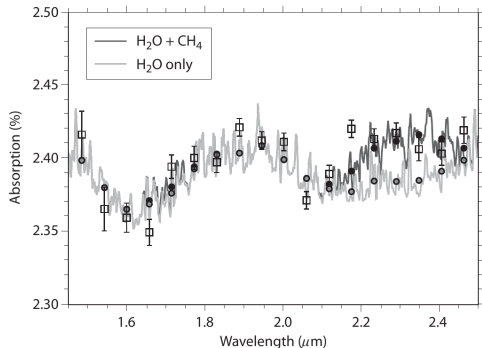


Figure 7: HST/NICMOS transmission spectrophotometry of HD 189733b (squares) with models containing H₂O only (gray) or H₂O + CH₄ (black). Seager (2010), Fig. 6.7.

- ▶ The signal is $\sim 5 \times 10^{-4}$ of the stellar flux, as estimated in Chapter 3: the area of an annulus about $5H$ tall relative to the stellar disk.
- ▶ Atoms and molecules detected in transmission in hot Jupiters include Na, K, H₂O, CO, CO₂ and (with JWST) SO₂.
- ▶ This particular CH₄ detection was later disputed: the NICMOS systematics had been underestimated (Gibson et al. 2011). A useful reminder that a feature is only as good as the instrument model behind it.

Seager's exercise 3(c) asks when thermal emission from the planet would matter in transmission: when the planet's own emission along the chord is not negligible compared with the transmitted starlight, i.e. for very hot planets observed in the infrared.

Thermal emission: the formal solution with $S = B$

At infrared wavelengths molecular absorption dominates and scattering can be neglected: $\kappa = \alpha$ (absorption only), and in LTE $S_{\text{therm}} = \varepsilon / \kappa = B(\tau_{\nu}, \nu)$. The equation $\mu \, dI / d\tau_{\nu} = I - B$ has the solution

$$I(0, \mu, \nu) = \int_0^{\infty} B(\tau_{\nu}, \nu) e^{-\tau_{\nu}/\mu} \frac{d\tau_{\nu}}{\mu}, \quad F(0, \nu) = 2\pi \int_0^1 \int_0^{\infty} B e^{-\tau_{\nu}/\mu} d\tau_{\nu} d\mu.$$

- ▶ “Powerfully simple” (Seager): each layer is a black body; the only hard ingredient is the opacity, which fixes $\tau_{\nu}(z)$.
- ▶ Exact in LTE without scattering, given $T(z)$. This makes it the workhorse for secondary-eclipse and directly imaged spectra.
- ▶ Thermal emission peaks near the Planck maximum: $\sim 0.8\text{--}1 \, \mu\text{m}$ for 2000–3000 K ultra-hot Jupiters, $\sim 15 \, \mu\text{m}$ for Earth, longer for colder planets.

An emission spectrum: HD 189733b

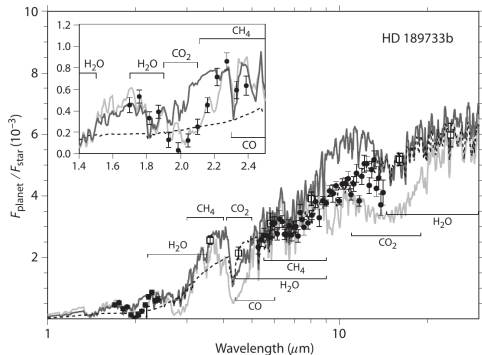


Figure 8: Secondary-eclipse data for HD 189733b: Spitzer photometry (squares), Spitzer/IRS (circles), HST/NICMOS (inset), with best-fit models. Seager (2010), Fig. 6.8.

- ▶ The planet/star contrast reaches $\sim 5 \times 10^{-3}$ in the mid-infrared, ten times the transmission signal.
- ▶ Order of magnitude (Chapter 3, Rayleigh–Jeans limit):

$$\frac{F_p}{F_*} \approx \frac{T_{b,p}}{T_{b,*}} \left(\frac{R_p}{R_*} \right)^2.$$

With $R_p/R_* = 0.155$ and $T_{b,*} \approx 5000$ K, a contrast of 3.4×10^{-3} at $8 \mu\text{m}$ implies $T_{b,p} \approx 700$ K; the full Planck ratio gives ≈ 1300 K, close to the measured dayside value.

- ▶ Absorption features of H_2O and CO/CO_2 : temperature decreasing with height on this dayside.

A constant source function

Take S independent of depth between an upper level ($\tau_{\min}$) and a lower one ($\tau_{\max}$); along a vertical ray (Seager Eq. 6.27),

$$I(\tau_{\min}) = I(\tau_{\max}) e^{-(\tau_{\max} - \tau_{\min})} + S \left[1 - e^{-(\tau_{\max} - \tau_{\min})} \right].$$

The first term is what survives of the light entering from below, the second what the slab emits along the path.

- **Optically thick slab**, $\tau_{\max} - \tau_{\min} \gg 1$: $I = S$. The incoming light is forgotten; only local conditions matter, and in LTE $I = S = B$: a featureless continuum. *Lines do not form at large optical depth.*
- **Optically thin top layer**, $\tau_{\min} = 0$, $\tau_{\max} < 1$, so $e^{-\tau} \approx 1 - \tau$:

$$I(0) = I(\tau_{\max}) - \tau_{\max} [I(\tau_{\max}) - S].$$

The layer *removes* light if the background is brighter than its source function, and *adds* light if it is fainter, in proportion to its optical depth $\tau_{\max}(\nu)$.

Absorption or emission: the four cases

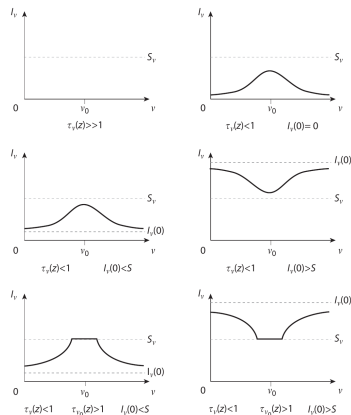


Figure 9: Line formation for a source function constant with depth and frequency. Seager (2010), Fig. 6.9, after Rutten.

Read the panels with $I(\tau_{\max}) \equiv I_\nu(0)$, the light from behind, and $\tau_\nu(z)$ the layer's optical depth, largest at line center ν_0 :

- **Top left** — thick at all ν : $I = S$, no line.
- **Top right** — thin, no backlight ($I_\nu(0) = 0$): emission line, strength $\propto \tau_\nu$.
- **Middle left** — thin, $I_\nu(0) < S$: emission line on top of the background.
- **Middle right** — thin, $I_\nu(0) > S$: absorption line.
- **Bottom** — thick at line center only ($\tau_{\nu_0} > 1$): the core saturates at $I = S$, giving a flat-topped emission or flat-bottomed absorption line.

Absorption lines require light from behind that is *brighter* than the layer's own source function — the same statement as “cool gas in front of a hot body”.

A linear source function: Eddington–Barbier revisited

Let $S(\tau_v) = a + b\tau_v$ (independent of μ). The formal solution gives exactly

$$I(0, \mu) = a + b\mu, \quad F(0) = 2\pi \int_0^1 (a + b\mu) \mu \, d\mu = \pi \left(a + \frac{2}{3}b \right) = \pi S(\tau_v = 2/3).$$

- ▶ The emergent **flux** is characteristic of $\tau_v \approx 2/3$; the **intensity** along the vertical ray, $I(0, 1) = S(\tau_v = 1)$. This is the Eddington–Barbier relation (Lecture 4).²
- ▶ It does not require the absence of scattering, only a source function that is (close to) linear in τ .
- ▶ Because τ_v depends on frequency, so does the geometric height of $\tau_v = 2/3$: this is the mathematical form of “the line core forms higher”.
- ▶ The same $I = a + b\mu$ gives $K = J/3$ for the second and zeroth moments — the **Eddington approximation**, which reappears as the third Eddington coefficient of the two-stream method.

²Seager writes $F(0, \nu) = S(\tau_v = 2/3)$ and $F(0, \nu) = S(\tau_v = 1)$ (Eqs. 6.32–6.33); the first lacks the factor π , and the second is the vertical intensity rather than a flux.

Why scattering makes everything harder

Absorption and thermal emission tie the radiation to the *local* temperature. Scattering does not:

- ▶ A scattered photon can travel far before it is absorbed. Scattering therefore **couples distant parts of the atmosphere**, and even the open boundary at the top influences the field deep inside.
- ▶ The source function now contains the radiation field itself,

$$S_{\text{scat}}(\tau, \mu, \phi) = \frac{\omega_0}{4\pi} \int_{4\pi} P(\mu, \phi; \mu', \phi') I(\tau, \mu', \phi') d\Omega',$$

so the transfer equation becomes an **integro-differential equation**: I appears on both sides, once differentiated and once integrated over all directions.

- ▶ Few analytic solutions exist; classical texts spend many pages even on isotropic scattering (Chandrasekhar 1960; Sobolev 1975).

Here ω_0 ($\tilde{\omega}$ in Seager) is the single-scattering albedo and P the single-scattering phase function, normalized as $\int_{4\pi} P d\Omega / 4\pi = 1$ (Lecture 4).

Direct beam and diffuse field

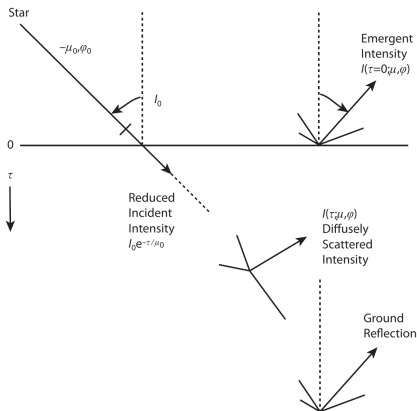


Figure 10: The attenuated direct beam from the star and the diffusely scattered intensity. Seager (2010), Fig. 6.10, after Chamberlain & Hunten.

Starlight arrives as a parallel beam from direction (μ_0, ϕ_0) :

$$I_*(0, \mu, \phi) = I_0 \delta(\mu - \mu_0) \delta(\phi - \phi_0).$$

Its flux through a horizontal surface is $\int \mu I_* d\Omega = \mu_0 I_0$: the incident flux is foreshortened by μ_0 (Chapter 2). Unscattered, the beam is attenuated as

$$I_{\text{dir}}(\tau) = I_0 e^{-\tau/\mu_0} \delta(\mu - \mu_0) \delta(\phi - \phi_0).$$

It is convenient to split the field into this **direct beam**, known analytically, and the **diffuse field** I of photons scattered at least once, which is what we must solve for.

The transfer equation with scattering

For the diffuse field, with the direct beam acting as a source,

$$\mu \frac{dI}{d\tau} = I - \underbrace{\frac{\omega_0}{4\pi} I_0 e^{-\tau/\mu_0} P(\mu, \phi; -\mu_0, \phi_0)}_{\text{single scattering of starlight}} - \underbrace{\frac{\omega_0}{4\pi} \int_{4\pi} P I d\Omega'}_{\text{multiple scattering}} - \underbrace{(1 - \omega_0) B}_{\text{thermal}}.$$

- ▶ The three source terms are the ones in Seager's exercise 7 and Fig. 6.12: singly scattered starlight, diffusely scattered radiation, and thermal emission; the fourth process, the attenuated direct beam, is I_{dir} itself.
- ▶ The sign of the scattering source is the same as that of any source, $\mu dI/d\tau = I - S$.³
- ▶ At visible wavelengths the thermal term vanishes; in the mid-infrared the scattering terms are usually small. The two-stream method of Heng keeps all of them.

³Seager's Eq. (6.38) prints $+S_{\text{scat}}$ on the right-hand side; the sign must be negative, as in Eq. (6.1).

Single scattering in a thin layer

In a layer of small optical depth most photons scatter at most once (on Earth: thin cirrus and aerosol layers). Keeping only the single-scattering source and integrating the formal solution,

$$I(0, \mu) = \frac{\omega_0 I_0 P}{4\pi\mu} \int_0^\tau e^{-\tau'/\mu_0} e^{-\tau'/\mu} d\tau' = \frac{\omega_0 \mu_0 I_0 P}{4\pi(\mu + \mu_0)} \left[1 - e^{-\tau\left(\frac{1}{\mu} + \frac{1}{\mu_0}\right)} \right].$$

The two exponentials are the attenuation on the way in (slant τ/μ_0) and on the way out (slant τ/μ).⁴ For $\tau \ll 1$ the bracket is $\tau(\mu + \mu_0)/(\mu\mu_0)$ and

$$R(\mu, \phi; \mu_0, \phi_0) \equiv \frac{\pi I(0, \mu, \phi)}{\mu_0 I_0} = \frac{\omega_0 \tau}{4\mu\mu_0} P(\mu, \phi; -\mu_0, \phi_0)$$

- R is the dimensionless **bidirectional reflectance**: the reflected intensity relative to that of a Lambert surface under the same illumination. Satellites measure it for Earth.
- In the thin limit it is *proportional to the optical depth* and maps the phase function directly: the brightness versus angle of a thin haze is its $P(\Theta)$.

⁴Seager's Eq. (6.45) prints the first exponential as $e^{-\tau\mu_0}$; it must be $e^{-\tau/\mu_0}$.

Multiple scattering and the scattered-light spectrum

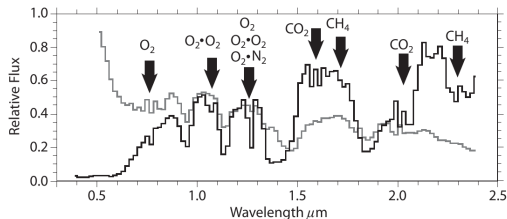


Figure 11: Earth as an exoplanet: hemispherically integrated scattered light from earthshine (gray) and the transmission spectrum measured during a lunar eclipse (black). Seager (2010), Fig. 6.11, from Pallé et al. (2009).

- ▶ No simple solution exists once photons scatter many times. The **adding-doubling** method builds the reflection and transmission of a thick layer from those of thin ones, doubling the thickness at each step; layers with different properties are then added.
- ▶ Hot Jupiters are *dark* ($A_g < 0.15$), compared with Jupiter's 0.5, so scattered light is much harder to detect than thermal emission.
- ▶ The two Earth spectra differ: transmission (long slant paths through the limb) shows O_2 , O_2-O_2 , O_2-N_2 collision bands and CO_2 , CH_4 more strongly than the disk-averaged reflected light, and a Rayleigh slope toward the blue.

What is the two-stream approximation?

Radiation in an atmosphere travels in all directions, is absorbed, re-emitted and scattered many times, possibly changing wavelength. Heng's strategy is to strip the problem down in two steps.

- ▶ **Physical basis.** In a *shallow* atmosphere (vertical extent $\ll R_p$) horizontal transport can be neglected, and radiation effectively flows only **up** and **down**. This holds for Earth, Mars, Titan and Venus, and reasonably for hot Jupiters, whose model domains are set by how deep starlight penetrates.
- ▶ **Mathematical basis.** The transfer equation has two independent variables, τ and μ . Integrating separately over the upward and the downward hemisphere removes μ and leaves two ordinary differential equations in τ alone.
- ▶ **The price.** Integration always produces one more unknown than equations. The system is closed by assuming ratios between angular moments — the **Eddington coefficients**. Different choices are different **closures**, and a careless choice can violate energy conservation.

Separate two-stream calculations are often done for starlight (shortwave) and for thermal emission (longwave), each with its own coefficients; this is the basis of the temperature–pressure profiles of the next lecture.

The starting equation and the moments

Heng's transfer equation with non-isotropic but **coherent** scattering (no change of wavelength), Eq. (3.1):

$$\mu \frac{\partial I}{\partial \tau} = I - \omega_0 \int_{4\pi} P I d\Omega' - (1 - \omega_0) B, \quad \int_{4\pi} P d\Omega = 1.$$

(Heng normalizes P to unity, so his P is our $P/4\pi$.) Define hemispheric moments of the intensity, with $d\Omega = d\mu d\phi$:

$$J_{\uparrow} = \int_0^{2\pi} \int_0^1 I d\mu d\phi, \quad J_{\downarrow} = \int_0^{2\pi} \int_{-1}^0 I d\mu d\phi, \quad F_{\uparrow} = \int_0^{2\pi} \int_0^1 \mu I d\mu d\phi, \quad F_{\downarrow} = \int_0^{2\pi} \int_{-1}^0 \mu I d\mu d\phi.$$

- ▶ $J = J_{\uparrow} + J_{\downarrow}$ is Heng's "total intensity", i.e. 4π times the *mean* intensity used by Seager.
- ▶ $F_{+} = F_{\uparrow} + F_{\downarrow}$ is the **total flux** and $F_{-} = F_{\uparrow} - F_{\downarrow}$ the **net flux**.⁵
- ▶ The analogy is with fluids: density and momentum are the moments of the Boltzmann equation.

⁵Although the integral for $F_{\downarrow}$ runs over $\mu < 0$, in the two-stream equations $F_{\downarrow}$ is used as a positive magnitude (an opaque isothermal layer gives $F_{\downarrow} = \pi B$); only then is $F_{-} = F_{\uparrow} - F_{\downarrow}$ the net upward energy flux.

Isotropic scattering: one equation, three unknowns

For isotropic scattering, $P = 1/4\pi$, the integral becomes $\omega_0 J/4\pi$:

$$\mu \frac{\partial I}{\partial \tau} = I - \frac{\omega_0 J}{4\pi} - (1 - \omega_0) B.$$

Introduce a slant optical depth $\tau' = \tau/\bar{\mu}_\uparrow$, with $\bar{\mu}_\uparrow$ a characteristic cosine of the upward hemisphere, and integrate over that hemisphere:

$$\frac{1}{\bar{\mu}_\uparrow} \frac{\partial F_\uparrow}{\partial \tau'} = J_\uparrow - \frac{\omega_0 J}{2} - 2\pi(1 - \omega_0) B.$$

- ▶ One equation, three unknowns ($F_\uparrow, J_\uparrow, J_\downarrow$). The downward hemisphere adds an equation but also $F_\downarrow$: the system stays *under-determined*.
- ▶ This is generic for moment methods: the equation for the n -th moment always contains the $(n+1)$ -th.
- ▶ The factor $\bar{\mu}$ is book-keeping: it keeps track of the minus signs in the downward hemisphere and drops out of the final solutions.

Closing the system: the Eddington coefficients

Assume the *ratios* of fluxes to total intensities are constants:

$$\epsilon_{\uparrow} \equiv \frac{F_{\uparrow}}{J_{\uparrow}}, \quad \epsilon_{\downarrow} \equiv \frac{F_{\downarrow}}{J_{\downarrow}}, \quad \epsilon \equiv \frac{F_{+}}{J}.$$

There is no physical reason for these ratios to be constant; it is a closure that makes progress possible.

Assuming the two hemispheres are symmetric, $\bar{\mu} = \bar{\mu}_{\uparrow} = -\bar{\mu}_{\downarrow}$ and $\epsilon' = \epsilon_{\uparrow} = \epsilon_{\downarrow}$, gives

$$\boxed{\frac{\partial F_{\uparrow}}{\partial \tau'} = \gamma_a F_{\uparrow} - \gamma_s F_{\downarrow} - \gamma_B B, \quad \frac{\partial F_{\downarrow}}{\partial \tau'} = -\gamma_a F_{\downarrow} + \gamma_s F_{\uparrow} + \gamma_B B}$$

$$\gamma_a = \bar{\mu} \left(\frac{1}{\epsilon'} - \frac{\omega_0}{2\epsilon} \right), \quad \gamma_s = \frac{\bar{\mu} \omega_0}{2\epsilon}, \quad \gamma_B = 2\pi \bar{\mu} (1 - \omega_0).$$

- ▶ Two coupled, first-order, linear ODEs with constant coefficients.
- ▶ γ_s is the **coupling**: scattering turns upward flux into downward flux and vice versa. Without scattering ($\gamma_s = 0$) the two streams are independent.

Fixing the coefficients with two physical limits

(1) Pure scattering must not heat. Radiative equilibrium (Lecture 4) requires $\partial F_- / \partial z = 0$. With $\omega_0 = 1$ ($\gamma_B = 0$), subtracting the two equations gives $\partial F_- / \partial \tau' = (\gamma_a - \gamma_s) F_+$. Since scattering is coherent, this must vanish at *each* wavelength, while $F_+ \neq 0$:

$$\gamma_a = \gamma_s \implies \varepsilon = \varepsilon'.$$

A single Eddington coefficient remains, the **first Eddington coefficient** ε .

(2) An opaque isothermal absorber must emit πB per hemisphere. For $\omega_0 = 0$ and $\Delta\tau \rightarrow \infty$ the solutions give $F_\uparrow = F_\downarrow = \gamma_B B / \gamma_a$. Evaluating $\int \mu B d\Omega$ over a hemisphere gives πB , so $\gamma_B / \gamma_a = \pi$ and

$$\boxed{\varepsilon = \frac{1}{2}} \quad \gamma_a = \bar{\mu} (2 - \omega_0), \quad \gamma_s = \bar{\mu} \omega_0, \quad \gamma_B = 2\pi \bar{\mu} (1 - \omega_0).$$

There is *no freedom* left: the coefficients and ε are not independent knobs, a frequent misconception.

The diffusivity factor

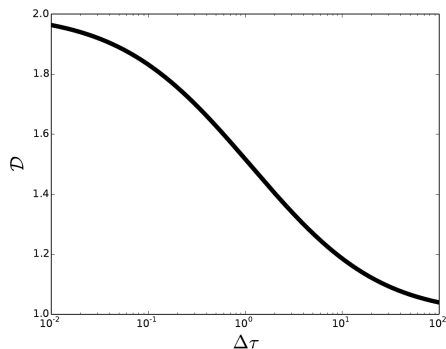


Figure 12: The diffusivity factor that makes the two-stream transmission exact for a purely absorbing isothermal layer. Heng (2017), Fig. 3.1.

In the purely absorbing isothermal case the two-stream transmission function is $\mathcal{T} = e^{-\Delta\tau/\varepsilon} = e^{-\mathcal{D}\Delta\tau}$, with the **diffusivity factor** $\mathcal{D} = 1/\varepsilon = 2$.

The exact transmission of diffuse light through a layer is $\mathcal{T} = 2E_3(\Delta\tau)$ (Lecture 4). Equating the two,

$$\mathcal{D} = -\frac{\ln[(1 - \Delta\tau)e^{-\Delta\tau} + \Delta\tau^2 E_1(\Delta\tau)]}{\Delta\tau} = -\frac{\ln 2E_3(\Delta\tau)}{\Delta\tau}.$$

- ▶ $\mathcal{D} \rightarrow 2$ for $\Delta\tau \ll 1$: the two-stream result is exact if the atmosphere is cut into optically thin layers.
- ▶ $\mathcal{D} \rightarrow 1$ for thick layers, where only nearly vertical rays get through.
- ▶ The popular $\mathcal{D} = 1.66$ was tuned for Earth. Heng's advice: do not treat $\mathcal{D}$ as a free parameter; use the value the closure implies.

General solution with isotropic scattering

Adding and subtracting the pair gives equations for the total and net fluxes:

$$\frac{\partial F_+}{\partial \tau'} = (\gamma_a + \gamma_s) F_-, \quad \frac{\partial F_-}{\partial \tau'} = (\gamma_a - \gamma_s) F_+ - 2\gamma_B B,$$

which combine into one second-order equation,

$$\frac{\partial^2 F_+}{\partial \tau'^2} - (\bar{\mu} \mathcal{D})^2 F_+ = -2\gamma_B (\gamma_a + \gamma_s) B, \quad \mathcal{D} = \frac{[(\gamma_a + \gamma_s)(\gamma_a - \gamma_s)]^{1/2}}{\bar{\mu}}.$$

- ▶ Within a layer the Planck function varies. Keep the first two terms of a Taylor series, $B = B_j + B'(\tau - \tau_j)$ with $B' \approx (B_2 - B_1)/(\tau_2 - \tau_1)$ — the linear source function again (Lecture 4).
- ▶ The solution is exponentials $e^{\pm \alpha \tau'}$ plus a particular part $\pi(B \pm B'/2)$:

$$F_{\uparrow} = A_1 \zeta_+ e^{\alpha \tau'} + A_2 \zeta_- e^{-\alpha \tau'} + \pi \left(B + \frac{B'}{2} \right), \quad F_{\downarrow} = A_1 \zeta_- e^{\alpha \tau'} + A_2 \zeta_+ e^{-\alpha \tau'} + \pi \left(B - \frac{B'}{2} \right),$$

with the **coupling coefficients** $\zeta_{\pm} = \frac{1}{2} \left[1 \pm (1 - \omega_0)^{1/2} \right]$.

- ▶ A_1, A_2 follow from the boundary conditions: $F_{\uparrow 2}$ entering from below and $F_{\downarrow 1}$ from above (layer 1 sits above layer 2).

The layer-to-layer solutions

Eliminating A_1, A_2 (Heng Eq. 3.37 and problem 3.8.2), the outgoing flux at the top of a layer is

$$F_{\uparrow 1} = \frac{1}{(\zeta_- \mathcal{T})^2 - \zeta_+^2} \left[(\zeta_-^2 - \zeta_+^2) \mathcal{T} F_{\uparrow 2} - \zeta_- \zeta_+ (1 - \mathcal{T}^2) F_{\downarrow 1} + (\text{Planck terms}) \right], \quad \mathcal{T} = e^{-\mathcal{D} \Delta \tau},$$

and $F_{\downarrow 2}$ follows by symmetry.

- ▶ Every non-flux quantity ($\mathcal{T}, \zeta_{\pm}$) is dimensionless and lies between 0 and 1: numerically well behaved.
- ▶ $\zeta_{\pm}$ act as *weights*: how much of the light entering from below is transmitted ($\propto \mathcal{T}$) and how much of the light entering from above is reflected back ($\propto \zeta_- \zeta_+$).
- ▶ A model atmosphere is a stack of layers with boundary conditions at the top (starlight) and bottom (interior heat). The solutions are applied to each pair of layers and iterated until radiative equilibrium is reached.

Pure absorption ($\zeta_+ = 1, \zeta_- = 0$): $F_{\uparrow 1} = \mathcal{T} F_{\uparrow 2} + \pi(B_{1+} - B_{2+} \mathcal{T})$. An opaque layer ($\mathcal{T} = 0$) shows only its own top black body; a transparent one ($\mathcal{T} = 1$) passes the boundary flux plus a term that exists only if the layer is not isothermal.

Test case: a scattering slab

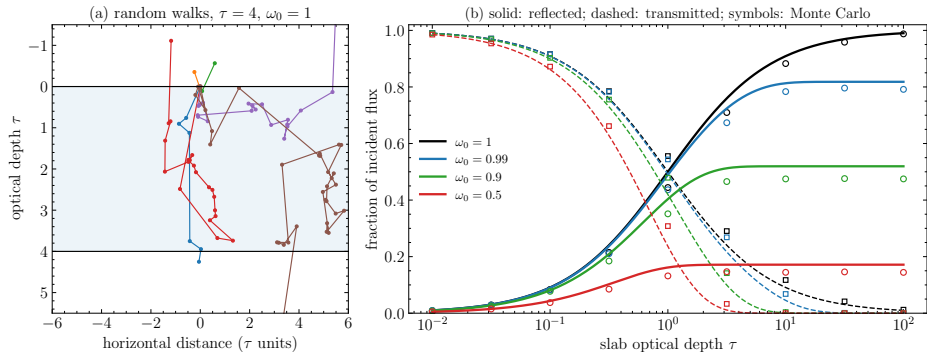


Figure 13: A homogeneous slab with isotropic scattering, lit from above by diffuse light, no emission. Lines: two-stream (hemispheric closure); symbols: Monte Carlo. Script `scripts/make_twostream_slab.py`.

Setting $B = 0$, $F_{\uparrow 2} = 0$: $R = \frac{\zeta_- \zeta_+ (1 - \mathcal{T}^2)}{\zeta_+^2 - \zeta_-^2 \mathcal{T}^2}$, $T_{\text{slab}} = \frac{(\zeta_+^2 - \zeta_-^2) \mathcal{T}}{\zeta_+^2 - \zeta_-^2 \mathcal{T}^2}$; a semi-infinite atmosphere reflects $R_\infty = \zeta_- / \zeta_+$.

What the test case teaches

- ▶ **Absorption caps the albedo.** However thick the slab, $R_\infty = \zeta_-/\zeta_+ = \frac{1 - \sqrt{1 - \omega_0}}{1 + \sqrt{1 - \omega_0}}$: 0.82 for $\omega_0 = 0.99$, 0.52 for 0.9, 0.17 for 0.5. Even 1% absorption per scattering removes a fifth of the light, because a reflected photon has scattered many times.
- ▶ **Conservative scattering** ($\omega_0 = 1$) needs its own solution (problem 3.8.4): $R = \tau/(1 + \tau)$, $T_{\text{slab}} = 1/(1 + \tau)$. Transmission falls only as $1/\tau$, not exponentially: photons diffuse through. The general formulas give 0/0 here — Heng’s caveat that “purely scattering” is not “purely reflecting”.
- ▶ **Accuracy.** Against the Monte Carlo result the two-stream values are off by $\sim 10\text{--}20\%$ at $\tau \sim 1$ (e.g. $\omega_0 = 1$: 0.50 versus 0.45). This is the typical price of two streams, and why multi-stream methods are used when scattering details matter.
- ▶ **Random walk.** In panel (a) photons wander far sideways before escaping; the number of steps grows as $\sim \tau^2$ (Lecture 4). That is the diffusion limit, discussed below.

The scattering phase function

Non-isotropic scattering can be handled *without* specifying P , using only two general properties.

1. P depends only on the **scattering angle** $\theta'' = \theta' - \theta$, with

$$\mu'' = \cos \theta'' = \mu' \mu + (1 - \mu'^2)^{1/2} (1 - \mu^2)^{1/2} \cos(\phi' - \phi).$$

μ'' is unchanged if both $\mu \rightarrow -\mu$ and $\mu' \rightarrow -\mu'$, so P is invariant under this double sign flip.

2. P integrates to one in any of the three angular frames: $\int P d\Omega = \int P d\Omega' = \int P d\Omega'' = 1$.

The usual fitting function is the **Heney–Greenstein** phase function,

$$P = \frac{1}{4\pi} \frac{1 - g_0^2}{(1 + g_0^2 - 2g_0 \cos \theta'')^{3/2}},$$

where g_0 is an input parameter; $g_0 = 0$ recovers $P = 1/4\pi$. In the derivation that follows, g_0 *emerges* as an integral over P .

Moments with non-isotropic scattering

Multiply Eq. (3.1) by a function $H(\theta)$ and integrate over all angles.

- $H = 1$ gives the net-flux equation,

$$\frac{\partial F_-}{\partial \tau'} = \bar{\mu} (1 - \omega_0)(J - 4\pi B),$$

already of the isotropic form: scattering changes directions, not the energy budget.

- $H = \mu$: writing $\mu = \cos(\theta' - \theta'')$, the cross term is odd under the double sign flip and vanishes; what remains is the first moment of the phase function,

$$g_0 \equiv \int_{4\pi} \mu'' P d\Omega'',$$

the **asymmetry factor** ($g_0 > 0$ forward, < 0 backward, 0 isotropic). It is to P what the flux is to I . With the second moments $K_{\uparrow,\downarrow} = \int \mu^2 I d\Omega$ over each hemisphere and $K_- = K_{\uparrow} - K_{\downarrow}$,

$$\frac{\partial K_-}{\partial \tau'} = \bar{\mu} F_+ (1 - \omega_0 g_0).$$

K/c is the radiation pressure.

The second Eddington coefficient and the hemispheric closure

Two equations, four unknowns (J, F_-, F_+, K_-). The first Eddington coefficient links J and F_+ ($\varepsilon = 1/2$); one more is needed. Define $\varepsilon_2 \equiv K_-/F_+$ and demand that the isotropic equations be recovered, which fixes $\varepsilon_2 = F_+/2F_-$ and gives

$$\frac{\partial F_+}{\partial \tau'} = 2\bar{\mu} F_- (1 - \omega_0 g_0).$$

Comparing with the isotropic pair yields the general coefficients

$$\gamma_a = \bar{\mu} [2 - \omega_0(1 + g_0)], \quad \gamma_s = \bar{\mu} \omega_0(1 - g_0), \quad \gamma_B = 2\pi\bar{\mu} (1 - \omega_0)$$

and coupling coefficients $\zeta_{\pm} = \frac{1}{2} [1 \pm \sqrt{(1 - \omega_0)/(1 - \omega_0 g_0)}]$.

- ▶ This is the **hemispheric** (hemi-isotropic) closure, here *derived* rather than assumed.
- ▶ The solutions keep the same form as for isotropic scattering, with $\mathcal{T} = e^{-\mathcal{D}\Delta\tau}$ and $\mathcal{D} = 2\sqrt{(1 - \omega_0)(1 - \omega_0 g_0)}$.
- ▶ Forward scattering acts like weaker scattering: γ_s depends on $\omega_0(1 - g_0)$, so a strongly forward-scattering cloud is more transparent than its ω_0 alone suggests.

Different closures

closure	$\gamma_a/\bar{\mu}$	$\gamma_s/\bar{\mu}$	$\gamma_B/\bar{\mu}$
hemispheric	$2 - \omega_0(1 + g_0)$	$\omega_0(1 - g_0)$	$2\pi(1 - \omega_0)$
Eddington	$\frac{1}{4}[7 - \omega_0(4 + 3g_0)]$	$-\frac{1}{4}[1 - \omega_0(4 - 3g_0)]$	$2\pi(1 - \omega_0)$
quadrature	$\frac{\sqrt{3}}{2}[2 - \omega_0(1 + g_0)]$	$\frac{\sqrt{3}}{2}\omega_0(1 - g_0)$	$\sqrt{3}\pi(1 - \omega_0)$

Two tests every closure must pass:

1. **Energy conservation for pure scattering:** $\partial F_-/\partial\tau \propto \gamma_a - \gamma_s \propto 1 - \omega_0$. All three pass.
2. **Correct black body for an opaque, isothermal absorber:** the figure of merit $f_\infty \equiv \gamma_B/[\pi(\gamma_a - \gamma_s)]$ must equal 1 (each hemisphere emits πB). Hemispheric and quadrature pass.

Why the Eddington closure fails the second test

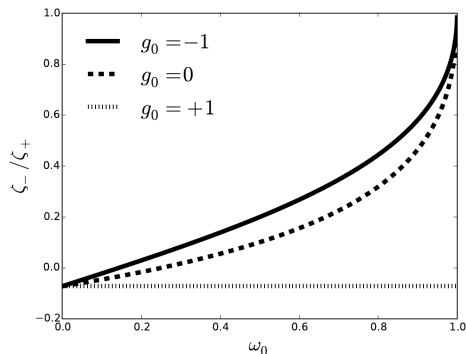


Figure 14: Spurious enhancement or reduction factor ζ_-/ζ_+ of the Eddington closure for an opaque layer. Heng (2017), Fig. 3.2.

- ▶ At $\omega_0 = 0$ (no scattering at all) the Eddington closure gives $\gamma_s = -\bar{\mu}/4 \neq 0$: upward rays are turned into downward ones, i.e. *reflection without scatterers*.

- ▶ Then $\zeta_{\pm} = \frac{1}{2} [1 \pm \sqrt{(\gamma_a - \gamma_s)/(\gamma_a + \gamma_s)}]$ gives $\zeta_-/\zeta_+ \approx -0.07$ and

$$f_{\infty} = \frac{\gamma_B}{\pi(\gamma_a - \gamma_s)} \left(1 - \frac{\zeta_-}{\zeta_+} \right) \approx 1.07,$$

a spurious $\sim 10\%$ excess of black-body flux.

- ▶ In general both reflected and emitted fluxes are distorted by ζ_-/ζ_+ , by amounts that depend on ω_0 and g_0 and compound from layer to layer.

Heng's point: a model meant to scan broad parameter space must be physically consistent everywhere, not merely tuned to work for one planet. (Goody & Yung's textbook uses the Eddington closure.)

Deep in the atmosphere: radiative diffusion

When the photon mean free path is much shorter than the scale height, conditions change little over one free path. Expand the source function about the local depth τ (Seager Eqs. 6.48–6.51):

$$S(\tau') \approx B(\tau) + \frac{dB}{d\tau}(\tau' - \tau) \quad \implies \quad I(\tau, \mu) \approx B(\tau) + \mu \frac{dB}{d\tau}.$$

- ▶ The field is almost isotropic and Planckian, with a small anisotropy $\propto \mu$ that carries the flux; as $\tau \rightarrow \infty$, $I = S = B$.
- ▶ Its moments give $K = J/3$ exactly — Heng's **third Eddington coefficient** $\varepsilon_3 = K/J = 1/3$ — and a flux

$$F = 2\pi \int_{-1}^1 \mu I d\mu = \frac{4\pi}{3} \frac{dB}{d\tau} \quad (\text{upward positive, } \tau \text{ increasing downward}).$$

- ▶ Here the two streams merge: net and total fluxes are no longer distinct, and the two-stream picture gives way to diffusion. Scattering is effectively isotropic ($g_0 = 0$) and $J = 4\pi B$.

The equation of radiative diffusion

With $d\tau = -\kappa dz$ (Seager: κ per unit length) and integrating over frequency,

$$F_{\nu} = -\frac{4\pi}{3\kappa_{\nu}} \frac{\partial B_{\nu}}{\partial T} \frac{dT}{dz} \quad \Rightarrow \quad \boxed{F = -\frac{16\sigma_{\text{SB}}T^3}{3\kappa_{\text{R}}} \frac{dT}{dz}}$$

where the frequency average that makes this exact is the **Rosseland mean** (Lecture 4),

$$\frac{1}{\kappa_{\text{R}}} = \frac{\int_0^{\infty} \kappa_{\nu}^{-1} (\partial B_{\nu} / \partial T) d\nu}{\int_0^{\infty} (\partial B_{\nu} / \partial T) d\nu}, \quad \int_0^{\infty} \frac{\partial B_{\nu}}{\partial T} d\nu = \frac{4\sigma_{\text{SB}}T^3}{\pi}.$$

- ▶ The form of Fourier's law for heat conduction: flux = conductivity $\times$ temperature gradient, with the **radiative conductivity** $16\sigma_{\text{SB}}T^3/3\kappa_{\text{R}}$. Heat flows down the temperature gradient.
- ▶ Heng uses opacity per unit mass: replace κ_{R} by $\rho \kappa_{\text{R}}$. With hydrostatic equilibrium, $dP = -\rho g dz$,

$$F = \frac{16\sigma_{\text{SB}}gT^3}{3\kappa_{\text{R}}} \frac{dT}{dP}.$$

Why a harmonic mean

Three ways to average 1 and 100 (the Pythagorean means):

$$\text{arithmetic } \frac{1+100}{2} = 50.5, \quad \text{geometric } (1 \times 100)^{1/2} = 10, \quad \text{harmonic } \frac{2}{1 + 1/100} \approx 2.$$

- ▶ The arithmetic mean weights both equally; the geometric mean averages in the logarithm; the harmonic mean is dominated by the *smallest* value.
- ▶ In an optically thick medium radiation leaks out through the **windows** — the wavelengths of lowest opacity — just as current in parallel resistors flows through the smallest resistance. The Rosseland mean is a harmonic mean for this reason.
- ▶ Hence a small amount of an opacity source that fills a window (a continuum absorber, a cloud) can change κ_R and the deep thermal structure far more than a strong line in an already opaque band.

Milne's solution

Deep in the atmosphere the flux is carried outward by radiation and, in radiative equilibrium, is constant with depth: set it to the internal heat flux, $F = \sigma_{\text{SB}} T_{\text{int}}^4$. Integrating $\sigma_{\text{SB}} T^4 = (16\sigma_{\text{SB}} T^3/3) dT/d\tau_R$ over the Rosseland optical depth τ_R ,

$$T^4 = \frac{3}{4} T_{\text{int}}^4 (\tau_R + C)$$

with $C \sim 1$ a constant of integration ($C = 2/3$ in the Eddington approximation).

- ▶ This is **Milne's solution**, first derived for stars. At $\tau_R \approx 1$ the observer measures $T \approx T_{\text{int}}$: for the Sun, $T_{\text{int}} \approx 5800$ K is the photospheric temperature.
- ▶ Temperature rises as $\tau_R^{1/4}$ with depth. In an irradiated planet this internal part is joined by the absorbed starlight; combining the two is the job of the next lecture.

Is the diffusion approximation valid? (Heng, problem 3.8.5)

Write the diffusion law as $F = -l_{\text{diff}} \partial(\sigma_{\text{SB}} T^4)/\partial z$, so $l_{\text{diff}} = 4/(3\rho \kappa_{\text{R}})$, and compare it with the scale height $H = k_{\text{B}}T/mg$.

- **Deep hot-Jupiter atmosphere:** $P = 100$ bar, $T = 1000$ K, $m = 2m_{\text{H}}$, $\kappa_{\text{R}} = 0.01 \text{ cm}^2 \text{ g}^{-1}$, $g = 10^3 \text{ cm s}^{-2}$. Then $\rho = Pm/k_{\text{B}}T = 2.4 \times 10^{-3} \text{ g cm}^{-3}$, $l_{\text{diff}} \approx 0.55 \text{ km}$, $H \approx 410 \text{ km}$, and $H/l_{\text{diff}} \approx 750$: **diffusion is excellent**.
- **Earth:** $P = 1$ bar gives a column mass $P/g \approx 1.0 \times 10^3 \text{ g cm}^{-2}$ (10 tonnes per m^2). An infrared optical depth of 2 implies $\kappa \approx 2 \times 10^{-3} \text{ cm}^2 \text{ g}^{-1}$. With $\rho = 1.2 \times 10^{-3} \text{ g cm}^{-3}$, $l_{\text{diff}} \approx 6 \text{ km}$ while $H \approx 8 \text{ km}$: $H/l_{\text{diff}} \approx 1.5$, so **diffusion fails**.

Earth's lower atmosphere is only marginally opaque in the infrared; its thermal structure needs full radiative transfer (plus convection), not a diffusion law.

Radiative transfer in cloudy atmospheres

The equations above apply to clouds too; the practical difficulty is the sharp jump in opacity at the edges of an optically thick cloud. Approaches in the literature:

- ▶ Put the lower boundary *at* the cloud top, arguing that photons scatter in the cloud until they are absorbed.
- ▶ Average the emergent intensity of clear and cloudy lines of sight (partial cloud cover, clouds at several heights).
- ▶ Replace the cloud edge by an exponential fall-off in density, for numerical convenience.
- ▶ Represent the cloud by an analytic reflection law instead of solving the transfer inside it.

For realistic, patchy, three-dimensional clouds the most accurate tool is **Monte Carlo** radiative transfer.

Monte Carlo: sampling the physics

Instead of solving for I , follow many individual photons and let random numbers decide what happens to each. The basic rule: a variable x with probability density $p(x)$ on $[a, b]$ is sampled from a uniform random number $\zeta \in [0, 1)$ by solving

$$\zeta = \int_a^{x_0} p(x) dx.$$

Example: how far does a photon travel? The probability of an interaction between τ and $\tau + d\tau$ is $p(\tau) d\tau = e^{-\tau} d\tau$, so

$$\zeta = 1 - e^{-\tau} \implies \boxed{\tau = -\ln(1 - \zeta)}$$

and the physical distance follows from the extinction coefficient along the path. In the same way one samples

- ▶ absorption versus scattering (absorbed if $\zeta > \omega_0$);
- ▶ the new direction from the phase function (isotropic: $\mu = 2\zeta - 1$, $\phi = 2\pi\zeta'$; Henyey–Greenstein has a closed inverse too);
- ▶ the entry direction (diffuse illumination: $\mu = \sqrt{\zeta}$).

The Monte Carlo algorithm

1. **Initialize:** launch a photon packet at the top of the atmosphere with a position and direction drawn from the stellar illumination.
 2. **Track:** draw the optical path to the next interaction and move the photon. Decide absorption or scattering. An absorbed photon is “killed”: its energy goes into heating and later thermal emission. A scattered photon gets a new direction (and polarization) from the phase function.
 3. **Escape:** repeat until the photon leaves the atmosphere; bin it by exit direction and position. The binned counts give the flux (and polarization) versus phase angle and inclination.
 4. **Errors:** counts in each bin obey Poisson statistics, so the relative error is $1/\sqrt{N}$; 10^7 or more photons are typical.
- **Strengths:** arbitrary geometry and inhomogeneity (patchy clouds, the full sphere, the terminator in transmission); intuitive.
 - **Weakness:** slow in optically thick media, where each photon needs $\sim \tau^2$ steps.

Our slab test used exactly this algorithm with 4×10^4 photons per point.

Putting all the pieces together

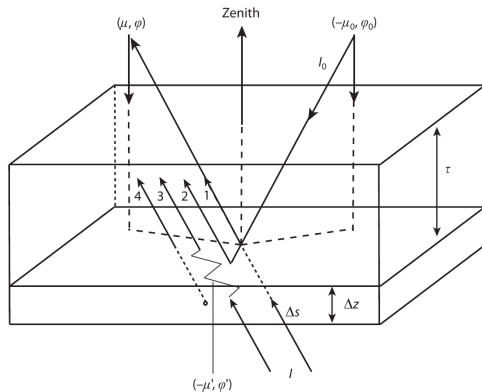


Figure 15: The four processes that change the intensity in a volume element: (1) attenuated starlight, (2) singly scattered starlight, (3) diffusely scattered radiation, (4) thermal emission. Seager (2010), Fig. 6.12, after Liou.

Seager's exercise 7: the full equation for the diffuse intensity,

$$\mu \frac{dI}{d\tau} = I - \frac{\omega_0}{4\pi} I_0 e^{-\tau/\mu_0} P(\mu, \phi; -\mu_0, \phi_0) - \frac{\omega_0}{4\pi} \int_{4\pi} P I d\Omega' - (1 - \omega_0) B.$$

Where each term dominates:

- (1) the direct beam: transit spectroscopy, thin atmospheres;
- (2) single scattering: thin hazes and aerosols ($\tau \ll 1$);
- (3) multiple scattering: thick clouds, Rayleigh-dominated visible light;
- (4) thermal emission: the infrared, where $\omega_0 \approx 0$.

Take-home messages

Line formation. Spectral features measure temperature *contrasts*: an isothermal LTE atmosphere emits a featureless black body. Opaque wavelengths form high, transparent ones deep; temperature falling outward gives absorption, an inversion gives emission. Emergent flux $\approx \pi S(\tau_v = 2/3)$.

Solutions. Transmission: Beer's law along the chord. Thermal emission: $F = 2\pi \int B E_2 d\tau$, exact without scattering. Scattering makes the equation integro-differential; thin layers give a reflectance $\propto \omega_0 \tau P / 4\mu\mu_0$.

Two-stream. Integrate over hemispheres and close with Eddington coefficients. Physics fixes them: $\epsilon = 1/2$ from the black-body limit, and the hemispheric closure $\gamma_a = \bar{\mu}[2 - \omega_0(1 + g_0)]$, $\gamma_s = \bar{\mu}\omega_0(1 - g_0)$. The Eddington closure makes reflection without scatterers. Accuracy ~ 10 – 20% .

Deep layers. Diffusion, $F = -(16\sigma_{\text{SB}}T^3/3\kappa_{\text{R}})dT/dz$, with the harmonic Rosseland mean; Milne's $T^4 = \frac{3}{4}T_{\text{int}}^4(\tau_R + C)$. Valid in hot-Jupiter interiors, not in Earth's troposphere.

Next. Temperature–pressure profiles: combining shortwave heating and longwave cooling in the two-stream framework (Heng Ch. 4).

References

- ▶ Seager, S. 2010, *Exoplanet Atmospheres: Physical Processes* (Princeton University Press), Chapter 6.
- ▶ Heng, K. 2017, *Exoplanetary Atmospheres: Theoretical Concepts and Foundations* (Princeton University Press), Chapter 3.
- ▶ Chandrasekhar, S. 1960, *Radiative Transfer* (Dover).
- ▶ Mihalas, D. 1978, *Stellar Atmospheres*, 2nd ed. (Freeman).
- ▶ Sobolev, V. V. 1975, *Light Scattering in Planetary Atmospheres* (Pergamon).
- ▶ Liou, K. N. 2002, *An Introduction to Atmospheric Radiation* (Academic Press); Goody, R. M. and Yung, Y. L. 1989, *Atmospheric Radiation* (Oxford).
- ▶ Toon, O. B., et al. 1989, *JGR* **94**, 16287; Meador, W. E. and Weaver, W. R. 1980, *J. Atmos. Sci.* **37**, 630 — two-stream closures.
- ▶ Heng, K., Mendonça, J. M. and Lee, J.-M. 2014, *ApJS* **215**, 4 — the two-stream solutions used here.
- ▶ Henyey, L. G. and Greenstein, J. L. 1941, *ApJ* **93**, 70.
- ▶ Madhusudhan, N. and Seager, S. 2009, *ApJ* **707**, 24; Line, M. R., et al. 2016, *AJ* **152**, 203; Evans, T. M., et al. 2017, *Nature* **548**, 58 — thermal inversions.
- ▶ Swain, M. R., et al. 2008, *Nature* **452**, 329; Gibson, N. P., et al. 2011, *MNRAS* **411**, 2199.
- ▶ Pallé, E., et al. 2009, *Nature* **459**, 814.
- ▶ Code, A. D. and Whitney, B. A. 1995, *ApJ* **441**, 400 — Monte Carlo scattering.